Rating a 3x3 Roster on Small Samples

Canada · FIBA 3x3 Women's Series · 2026 club season

242 rated players · 598 club matches · 3,353 appearances

Source: Strong Side Analytics

Abstract

In a 3x3 basketball setting, the nature of the game provides very little data. A match lasts ten minutes and a roster holds four players, so a single appearance yields about nine possessions and five shot attempts. Across a full 2026 club season the median player in this league attempted 16 shots from behind the arc and four free throws. Raw shooting percentages are useless at that volume: sort a leaderboard by free-throw percentage and it will be led by someone who went two-for-two in April. 242 women's club players from the 2026 FIBA 3x3 Women's Series were rated on ten individual skills and three play types. Each rating blends a player's own record with a prior estimated from the league, so a thin sample gets pulled toward the league average instead of reaching the extremes. The result is then placed against the spread of talent the league actually shows. Placing it against the spread of the adjusted values would be circular, since shrinkage is what compressed those values in the first place. Each player's games were split in half and one half predicted from the other. Out-of-sample error rises monotonically as stated confidence falls, from 0.044 to 0.139, and the 90% predictive interval contains the held-out result between 90.2% and 95.0% of the time. Two skills fail. Free-throw and arc shooting cannot separate most of the league at any point in a season: a confident rating needs roughly 24 free-throw and 41 arc attempts, and the median player takes four and sixteen respectively in a whole season. For those players the rating is largely the league average, and two players of genuinely different ability will receive near-identical numbers. Those ratings are still reported, with the limitation stated beside them.

1 Method

1.1 Data

Strong Side Analytics, 2026 season, women's club competition only. The sample is 598 matches and 3,353 player appearances; a player qualifies for a rating at five appearances or more.

1.2 Adjusting for sample size

A prior describes how a skill is spread across the league: where players cluster, and how far apart they sit. Two families are used here, each matched to the shape of the quantity it describes.

A Beta prior fits a proportion. Successes out of attempts cannot fall below zero or rise above one, and shooting works that way. Fitted to arc shooting, the Beta centres on 23.5%, puts the genuine gap between players at a few percentage points, and carries the weight of about 126 attempts.

A Gamma prior fits a rate: events per unit of opportunity, with no upper bound. Assists, rebounds, steals and turnovers are counted per possession and can take any non-negative value, so a proportion would be the wrong shape. Fitted to assists, the Gamma centres on 0.16 per possession and weighs about 39 possessions.

Each family is conjugate to the process that produced the data, Beta with makes over attempts and Gamma with counts over possessions. Conjugacy keeps the updated belief inside the same family, which makes the update arithmetic. What comes out is a weighted average of the player's own rate and the league rate, weighted by her sample against the prior's strength. A player with 126 arc attempts is half her own record and half the league. A player with 16 is mostly the league.

The priors are fitted to this league by maximum likelihood. Method of moments came first and was abandoned. It estimates the spread of true talent as observed variance minus sampling variance, a difference between two similar quantities, and the answer would not hold still: the fitted prior for arc shooting ranged from 91 to 230 attempts depending on nothing but the minimum-attempt cutoff. Maximum likelihood weights each player by her own sample, needs no cutoff, and returned 100 to 124 attempts across every setting tried.

A Wilson interval is the standard recommendation for this problem, and it shrinks toward 50%. For product ratings that is reasonable. For basketball it is wrong. Nobody in this league shoots 50% from behind the arc, where the mean is 23.5%, so a player who misses her first three attempts should be pulled down toward 23.5%.

1.3 The rating scale

The first approach ranked players by percentile of the adjusted values. On any skill the data cannot resolve, that fails. Almost nobody attempts enough free throws, so every estimate sits close to the prior and the league's adjusted values span just 63.0% to 73.2%, a standard deviation of 1.7 points. Percentile stretches that sliver across the whole scale. Katherine Plouffe, six points below the median, came out at 1 of 100.

Ratings are placed instead against the prior's own standard deviation, which is what the league genuinely varies by. A skill the data cannot resolve now lands near 50. Plouffe's free-throw rating moved from 1 to 14. Defensive rebounding has real spread and barely moved: Merissah Russell went from 19 to 24, still a below-average defensive rebounder on a large, well-measured sample.

1.4 Confidence

A conjugate posterior mean is a weighted average of the league rate and the player's own rate. That weight is reported directly, as the share of the rating coming from her own record; it is exact arithmetic rather than a rule of thumb. Beside it sits the highest share any player in the league reaches for that skill, because the two numbers read very differently. 31% own data on arc shooting sounds thin. The best-covered player in the league reaches 57%, and nobody clears 44% on free throws.

1.5 Validation

Every rating was rebuilt from each player's odd-numbered games and tested against her even-numbered ones. Mean absolute error rises as stated confidence falls, from 0.044 at very high to 0.139 at very low, so the labels rank trustworthiness in the right order. Shrinkage earns its keep where confidence is lowest, beating the raw rate by a factor of 1.48 at very low against 1.10 at very high.

Interval coverage needed a correction. Approximating the Beta and Gamma quantiles with a normal distribution produced intervals that held the held-out result as little as 35% of the time against a nominal 90%, and the failure was worst in the low-count regime where these posteriors are most skewed. Exact quantiles by continued fraction replaced the approximation. The reported interval is also predictive, carrying the noise of the future sample as well as uncertainty about the rate. Coverage now runs 90.2% to 95.0% across every confidence band.

1.6 What is left out

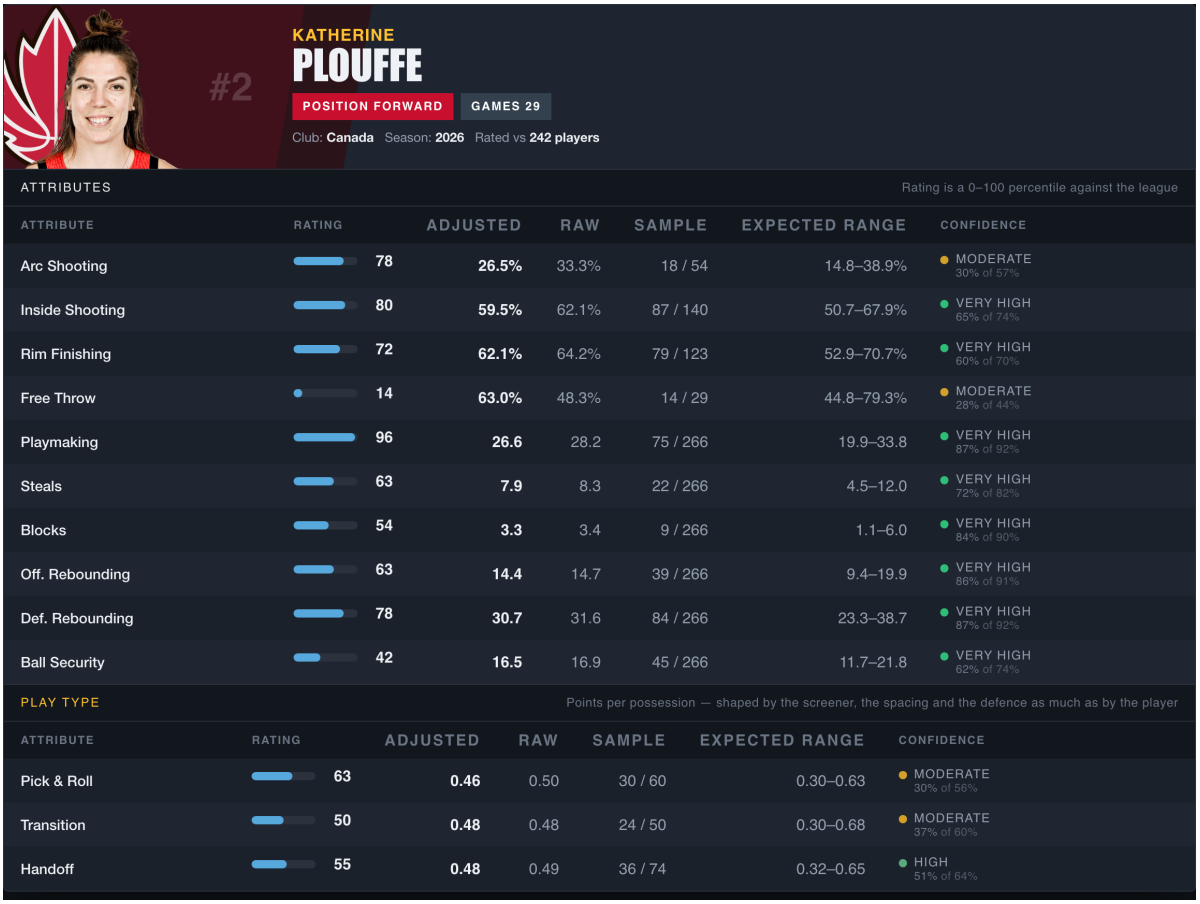
Seven of the ten tracked play types are excluded. Isolation, cut, spot up, post up, off-screen and offensive rebound average four to nine possessions per player per season, against 20 for pick and roll. The three retained account for 60% of all offence in the league. Shot-detail breakdowns go for the same reason: layups support a rating for 279 players, floaters for one, and corner threes for none.

There is no overall rating. One was built, tested against on-court net rating, and removed. A confidence-weighted mean of the attributes correlated at +0.41, a plain mean at +0.60, and weights fitted to net rating reached +0.68 under cross-validation. The fitted weights explain the weak averages: inside and arc shooting carry almost all the signal, while offensive and defensive rebounding contribute close to nothing. Averaging

those together dilutes what drives scoring margin. A single number could be defended, though an average cannot, and a fitted composite is not something a coach can recompute or argue with.

2 Results

Ratings run 0 to 100 against the league, where 50 is average and a move of 34 points corresponds to one standard deviation of true talent. Every row carries the sample it rests on, the expected range for a future sample of the same size, and the share of the rating that is the player's own record. Play types sit in their own block, because points per possession there reflects the screener, the spacing and the defence as much as the player.



KATHERINE PLOUFFE #2

POSITION FORWARD | GAMES 29

Club: Canada Season: 2026 Rated vs 242 players

ATTRIBUTES — Rating is a 0–100 percentile against the league

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Arc Shooting	78	26.5%	33.3%	18 / 54	14.8–38.9%	MODERATE 30% of 57%
Inside Shooting	80	59.5%	62.1%	87 / 140	50.7–67.9%	VERY HIGH 65% of 74%
Rim Finishing	72	62.1%	64.2%	79 / 123	52.9–70.7%	VERY HIGH 60% of 70%
Free Throw	14	63.0%	48.3%	14 / 29	44.8–79.3%	MODERATE 28% of 44%
Playmaking	96	26.6	28.2	75 / 266	19.9–33.8	VERY HIGH 87% of 92%
Steals	63	7.9	8.3	22 / 266	4.5–12.0	VERY HIGH 72% of 82%
Blocks	54	3.3	3.4	9 / 266	1.1–6.0	VERY HIGH 84% of 90%
Off. Rebounding	63	14.4	14.7	39 / 266	9.4–19.9	VERY HIGH 86% of 91%
Def. Rebounding	78	30.7	31.6	84 / 266	23.3–38.7	VERY HIGH 87% of 92%
Ball Security	42	16.5	16.9	45 / 266	11.7–21.8	VERY HIGH 62% of 74%

PLAY TYPE — Points per possession — shaped by the screener, the spacing and the defence as much as by the player

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Pick & Roll	63	0.46	0.50	30 / 60	0.30–0.63	MODERATE 30% of 56%
Transition	50	0.48	0.48	24 / 50	0.30–0.68	MODERATE 37% of 60%
Handoff	55	0.48	0.49	36 / 74	0.32–0.65	HIGH 51% of 64%



#7

PAIGE
CROZON

POSITION GUARD | GAMES 29

Club: **Canada** Season: **2026** Rated vs **242 players**

ATTRIBUTES — Rating is a 0–100 percentile against the league

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Arc Shooting	77	26.3%	29.7%	30 / 101	17.8–35.6%	HIGH 44% of 57%
Inside Shooting	66	56.9%	58.3%	70 / 120	47.5–66.7%	VERY HIGH 62% of 74%
Rim Finishing	64	60.8%	62.3%	66 / 106	50.9–70.8%	HIGH 57% of 70%
Free Throw	72	72.0%	80.8%	21 / 26	53.8–88.5%	MODERATE 26% of 44%
Playmaking	70	19.2	19.7	52 / 264	13.3–25.4	VERY HIGH 87% of 92%
Steals	64	8.0	8.3	22 / 264	4.5–12.1	VERY HIGH 72% of 82%
Blocks	74	4.6	4.9	13 / 264	1.9–8.0	VERY HIGH 84% of 90%
Off. Rebounding	80	17.1	17.8	47 / 264	11.7–23.1	VERY HIGH 86% of 91%
Def. Rebounding	52	24.9	25.0	66 / 264	18.2–32.2	VERY HIGH 87% of 92%
Ball Security	87	12.4	10.2	27 / 264	8.0–17.1	VERY HIGH 62% of 74%

PLAY TYPE — Points per possession — shaped by the screener, the spacing and the defence as much as by the player

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Pick & Roll	49	0.44	0.43	23 / 53	0.28–0.60	MODERATE 27% of 56%
Transition	76	0.53	0.60	43 / 72	0.38–0.71	HIGH 46% of 60%
Handoff	56	0.48	0.49	33 / 67	0.31–0.66	HIGH 48% of 64%



#13

MERISSAH
MELANIE RUSSELL

POSITION GUARD | GAMES 26

Club: **Canada** Season: **2026** Rated vs **242 players**

ATTRIBUTES — Rating is a 0–100 percentile against the league

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Arc Shooting	81	26.8%	33.3%	21 / 63	15.9–38.1%	MODERATE 33% of 57%
Inside Shooting	62	56.4%	57.5%	65 / 113	46.9–66.4%	VERY HIGH 60% of 74%
Rim Finishing	64	60.9%	62.6%	57 / 91	50.5–71.4%	HIGH 53% of 70%
Free Throw	55	69.5%	71.0%	22 / 31	54.8–83.9%	MODERATE 30% of 44%
Playmaking	64	18.2	18.6	42 / 226	12.4–24.8	VERY HIGH 85% of 92%
Steals	96	11.7	13.7	31 / 226	7.1–16.8	VERY HIGH 69% of 82%
Blocks	51	3.1	3.1	7 / 226	0.9–5.8	VERY HIGH 82% of 90%
Off. Rebounding	57	13.5	13.7	31 / 226	8.4–19.5	VERY HIGH 84% of 91%
Def. Rebounding	24	19.1	18.1	41 / 226	12.8–26.1	VERY HIGH 85% of 92%
Ball Security	79	13.3	11.5	26 / 226	8.4–18.6	HIGH 58% of 74%

PLAY TYPE — Points per possession — shaped by the screener, the spacing and the defence as much as by the player

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Pick & Roll	56	0.45	0.47	28 / 60	0.30–0.62	MODERATE 30% of 56%
Transition	82	0.55	0.68	30 / 44	0.34–0.77	MODERATE 34% of 60%
Handoff	76	0.52	0.60	34 / 57	0.35–0.72	HIGH 44% of 64%



#22

CASSANDRA
BROWN

POSITION GUARD | GAMES 16

Club: **Canada** Season: **2026** Rated vs **242 players**

ATTRIBUTES — Rating is a 0–100 percentile against the league

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Arc Shooting	60	24.5%	27.8%	10 / 36	11.1–38.9%	LOW 22% of 57%
Inside Shooting	79	59.3%	65.0%	39 / 60	46.7–71.7%	HIGH 45% of 74%
Rim Finishing	89	65.6%	77.8%	35 / 45	51.1–77.8%	MODERATE 36% of 70%
Free Throw	63	70.7%	87.5%	7 / 8	37.5–100.0%	VERY LOW 10% of 44%
Playmaking	49	15.6	15.5	18 / 116	8.6–24.1	VERY HIGH 75% of 92%
Steals	49	7.0	6.9	8 / 116	2.6–12.1	HIGH 53% of 82%
Blocks	64	3.9	4.3	5 / 116	0.9–8.6	VERY HIGH 69% of 90%
Off. Rebounding	47	12.2	12.1	14 / 116	6.0–19.8	VERY HIGH 72% of 91%
Def. Rebounding	73	29.4	31.0	36 / 116	19.0–40.5	VERY HIGH 74% of 92%
Ball Security	77	13.6	10.3	12 / 116	7.8–20.7	HIGH 42% of 74%

PLAY TYPE — Points per possession — shaped by the screener, the spacing and the defence as much as by the player

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Pick & Roll	55	0.44	0.50	9 / 18	0.17–0.72	VERY LOW 11% of 56%
Transition	87	0.57	0.82	23 / 28	0.32–0.86	LOW 25% of 60%
Handoff	62	0.49	0.56	14 / 25	0.24–0.76	MODERATE 26% of 64%



#1

MICHELLE
PLOUFFE

POSITION FORWARD | GAMES 8

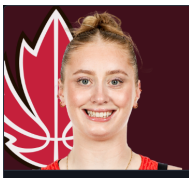
Club: **Canada** Season: **2026** Rated vs **242 players**

ATTRIBUTES — Rating is a 0–100 percentile against the league

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Arc Shooting	58	24.2%	28.6%	6 / 21	9.5–42.9%	LOW 14% of 57%
Inside Shooting	63	56.6%	61.3%	19 / 31	38.7–74.2%	MODERATE 30% of 74%
Rim Finishing	52	59.2%	60.0%	18 / 30	43.3–76.7%	MODERATE 27% of 70%
Free Throw	63	70.7%	87.5%	7 / 8	37.5–100.0%	VERY LOW 10% of 44%
Playmaking	91	24.2	29.0	20 / 69	13.0–37.7	VERY HIGH 64% of 92%
Steals	94	11.2	17.4	12 / 69	4.3–20.3	HIGH 40% of 82%
Blocks	62	3.8	4.3	3 / 69	0.0–8.7	HIGH 57% of 90%
Off. Rebounding	52	12.9	13.0	9 / 69	4.3–23.2	VERY HIGH 61% of 91%
Def. Rebounding	24	19.1	15.9	11 / 69	8.7–30.4	VERY HIGH 63% of 92%
Ball Security	50	15.9	15.9	11 / 69	7.2–26.1	MODERATE 30% of 74%

PLAY TYPE — Points per possession — shaped by the screener, the spacing and the defence as much as by the player

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Pick & Roll	54	0.44	0.48	11 / 23	0.22–0.70	LOW 14% of 56%
Transition	77	0.54	0.76	16 / 21	0.29–0.86	LOW 20% of 60%
Handoff	45	0.46	0.33	2 / 6	0.00–1.00	VERY LOW 8% of 64%



#8

TARA
WALLACK

POSITION FORWARD | GAMES 8

Club: **Canada** Season: **2026** Rated vs **242 players**

ATTRIBUTES

Rating is a 0–100 percentile against the league

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Arc Shooting	49	**23.4%**	22.2%	2 / 9	0.0–44.4%	VERY LOW 7% of 57%
Inside Shooting	52	**54.9%**	55.6%	20 / 36	38.9–69.4%	MODERATE 33% of 74%
Rim Finishing	46	**58.4%**	57.1%	20 / 35	42.9–74.3%	MODERATE 30% of 70%
Free Throw	40	**67.5%**	50.0%	3 / 6	33.3–100.0%	VERY LOW 8% of 44%
Playmaking	30	**12.4**	10.2	6 / 59	3.4–23.7	VERY HIGH 60% of 92%
Steals	48	**7.0**	6.8	4 / 59	1.7–13.6	MODERATE 37% of 82%
Blocks	67	**4.1**	5.1	3 / 59	0.0–10.2	HIGH 54% of 90%
Off. Rebounding	92	**19.9**	25.4	15 / 59	8.5–32.2	HIGH 57% of 91%
Def. Rebounding	53	**25.1**	25.4	15 / 59	11.9–39.0	HIGH 60% of 92%
Ball Security	52	**15.7**	15.2	9 / 59	6.8–25.4	MODERATE 27% of 74%

PLAY TYPE

Points per possession — shaped by the screener, the spacing and the defence as much as by the player

ATTRIBUTE	RATING	ADJUSTED	RAW	SAMPLE	EXPECTED RANGE	CONFIDENCE
Pick & Roll	40	**0.42**	0.29	4 / 14	0.14–0.71	VERY LOW 9% of 56%
Transition	45	**0.47**	0.40	4 / 10	0.10–0.90	VERY LOW 10% of 60%
Handoff	49	**0.47**	0.45	5 / 11	0.18–0.82	LOW 13% of 64%

3 Limitations

Play-type and rim-finishing figures come from SSA's season aggregates. This database holds no per-match rows for them, so they could not be split-half validated the way the other attributes were. The shrinkage and the confidence arithmetic are identical; the calibration is unverified.

Free-throw ratings will not improve. At 0.9 attempts per game, the attempts needed for a confident rating exceed a full season, and that is a property of the format rather than of the data collection. Arc shooting is the same for all but the highest-volume shooters.

Club and national-team results are not combined. A player's club rating says nothing about her national-team form, and the two competitions differ enough in quality that pooling them would mislead.

These are descriptions of what happened, not projections. A rating estimates a player's rate over the season observed, adjusted for how much of that season was actually seen.

References

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